

SUKET RAJ

Third Year Undergraduate

Department of Computer Science and Engineering, IIT Kanpur

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raj-suket 📧 | suket-raj 🌐

ACADEMIC QUALIFICATIONS

Year	Educational Qualification	Institute	CPI/%
2020 - Present	B.Tech in Computer Science and Engineering	Indian Institute of Technology Kanpur	8.91/10.0
2020	Class - XII (CBSE)	St. Paul Secondary School, Samastipur	93.0%
2018	Class - X (CBSE)	Jawahar Navodaya Vidyalaya Samastipur	93.4%

SCHOLASTIC ACHIEVEMENTS

- Awarded with an **A*** grade for **Exceptional Performance** in the course of Data Structures and Algorithms *2021*
- Secured **All India Rank 963** in Joint Entrance Examination (JEE)-Advanced-2020 among 0.25 million candidates *2020*
- Recipient of **Samsung Star Scholarship**, Samsung India (awarded to **top 110** students all over India) *2020*
- Secured **All India Rank 837**, Kishore Vaigyanik Protsahan Yojana (**KVPY**) **SA**, among 150,000 candidates *2019*
- National Top 1%**, National Science Examination in Physics (NSEP), Indian Association of Physics Teachers *2019*
- National Top 1%**, National Science Examination in Chemistry (NSEC), Indian Association of Chemistry Teachers *2019*
- Recipient of **NTSE Scholarship**, NCERT, Government of India (among **top 0.1%** students all over India) *2018*

WORK EXPERIENCE

Summer Intern | [needl.ai](#) 📧

May'22 - Jul'22

- Worked on data pipelining based on crawlers and user preferences to get the data to create a robust unified platform
- Implemented **integrated application system** using APIs of various applications such as OneNote, Twitter, Dropbox
- Tasked with and implemented the entire **authentication flow** for twitter application using the **passport strategy**
- Implemented automatic data uploading program and helped team members in automating other important processes

PROJECTS

Exec 📧 | *Course Project (CS253), Prof. Indranil Saha*

Jan'22 - Apr'22

- Collaborated with a team to build a web platform for **Election Conduction** using **springboot** framework in backend
- Contributed in 30+ **User classes** and **APIs** like Aspiring-Candidate user and Add Campaigner, tested it using Postman
- The platform can be used to digitalise election related activities like campaigning, reporting candidates, imposing fines

Game of Blocks 📧 | *Programming Club, IIT Kanpur*

May'21 - Jul'21

- Learnt the basics of blockchain, hashing, game theory, **school choice** and **voting** mechanisms and their uses in real life
- Worked in a team of three to implement **smart contracts** to solve real life situations such as election-voting, asset transfer
- Implemented **Two-Round-System** voting mechanism and **Top-Trading-Cycle** School-Choice Mechanism in **Solidity**

SAT Solver 📧 | *Course Project (CS202), Prof. Shubhajit Roy*

Jan'22 - Feb'22

- Implemented the **DPLL** algorithm in C++ to create a SAT solver and used various heuristics to break the CNF formula
- Designed a **Sudoku Generator and Solver** which generates CNF encodings that are then solved by the SAT solver

Algo101x | *Stamatics, IIT Kanpur*

Apr'22 - Jul'22

- Started with basic topics like CRT(**chinese remainder theorem**), **matrix exponentiation** and applications of GCD
- Explored some interesting algorithms like **bitmasking** and **dynamic programming**, **hashing** and **mobius inversion**

SPORT PROGRAMMING

- 5 Star on CodeChef** 📧 (highest rating **2096**) | **Expert on Codeforces** 📧 (highest rating **1779**)
- Achieved **Global Rank 16**, CodeChef July Cook-Off 2022 Div 2 | **Global Rank 254**, Codeforces Round 794 (Div. 2)
- Ranked **950**, **Google Kick Start 2021** Round C | Scored **1649905**, **Google Hash Code 2022** Qualification Round
- Awarded with **Sport Programmer of the Month**, a competition held for entire second Year students *Aug'21*
- Top 10, Fresher's Programming Contest 2020**, a competition held for all UG and PG freshmen students *Jan'21*

TECHNICAL SKILLS

- Programming Languages** : C/C++, Python, Java, Bash, Solidity, Verilog
- Libraries, Software & Utilities** : Git, Linux, Vim, $\text{\LaTeX}$, NumPy, Pandas, PySAT, Tensorflow

MISCELLANEOUS

Project Mentor | **Demystifying Algorithms, Programming Club, IIT Kanpur**

Jun'22 - Present

- Mentoring a group of students interested in advanced algorithms on topics like **Effective Searches** and **Heavy Hitters**

Speaker | [needl.ai](#) 📧

Jun'22

- Delivered a **Tech Talk** on **Trending Searches**, shared my knowledge about algorithms for extracting top searches

POSITIONS OF RESPONSIBILITY

Academic Mentor, Counselling Service, IIT Kanpur

Nov'21 - Jun'22

- Mentored freshmen UG students in the **Fundamentals of Computing** course, helped them solve other academic issues

Student Guide, Counselling Service, IIT Kanpur

Nov'21 - Present

- Guiding 5 freshmen emotionally and academically to help them get familiar with and transition into college environment

RELEVANT COURSEWORK

* : Online certified courses

Data Structures and Algorithms	Fundamentals of Computing	Software Development and Operations
Computer Organization	Probability for Computer Science	Discrete Mathematics
Logic for Computer Science	Linear Algebra and ODE	Algorithms on Graphs*